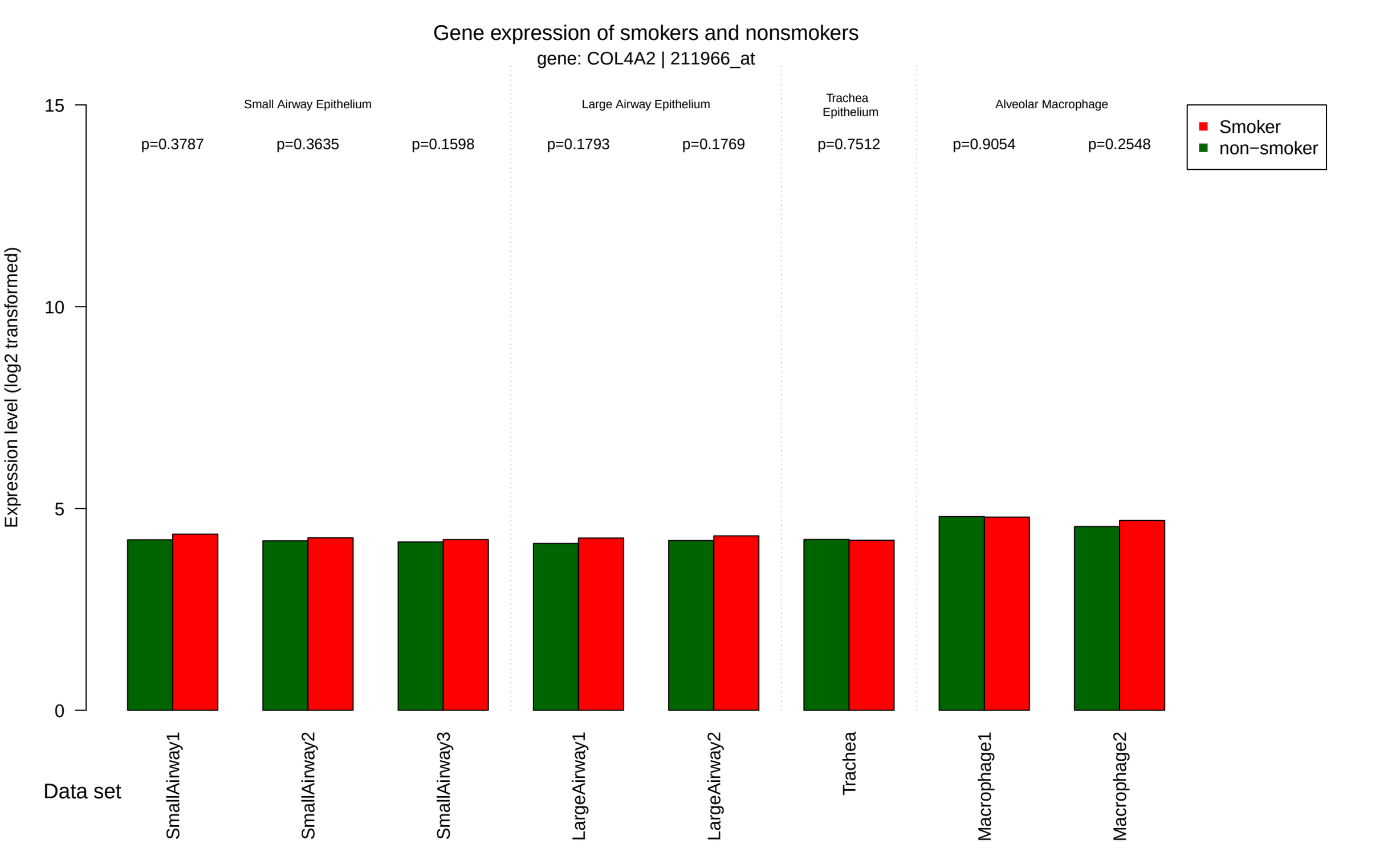
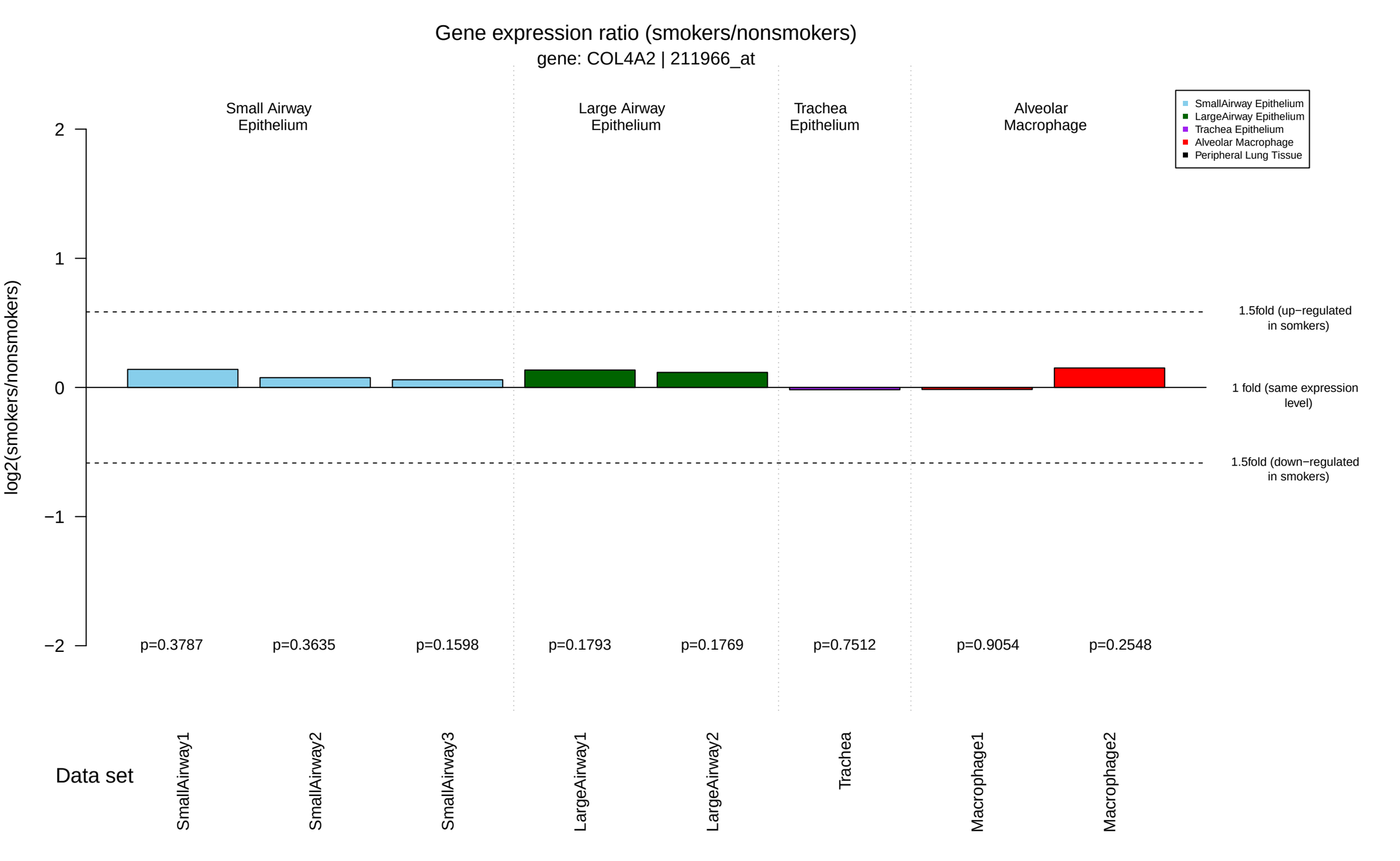
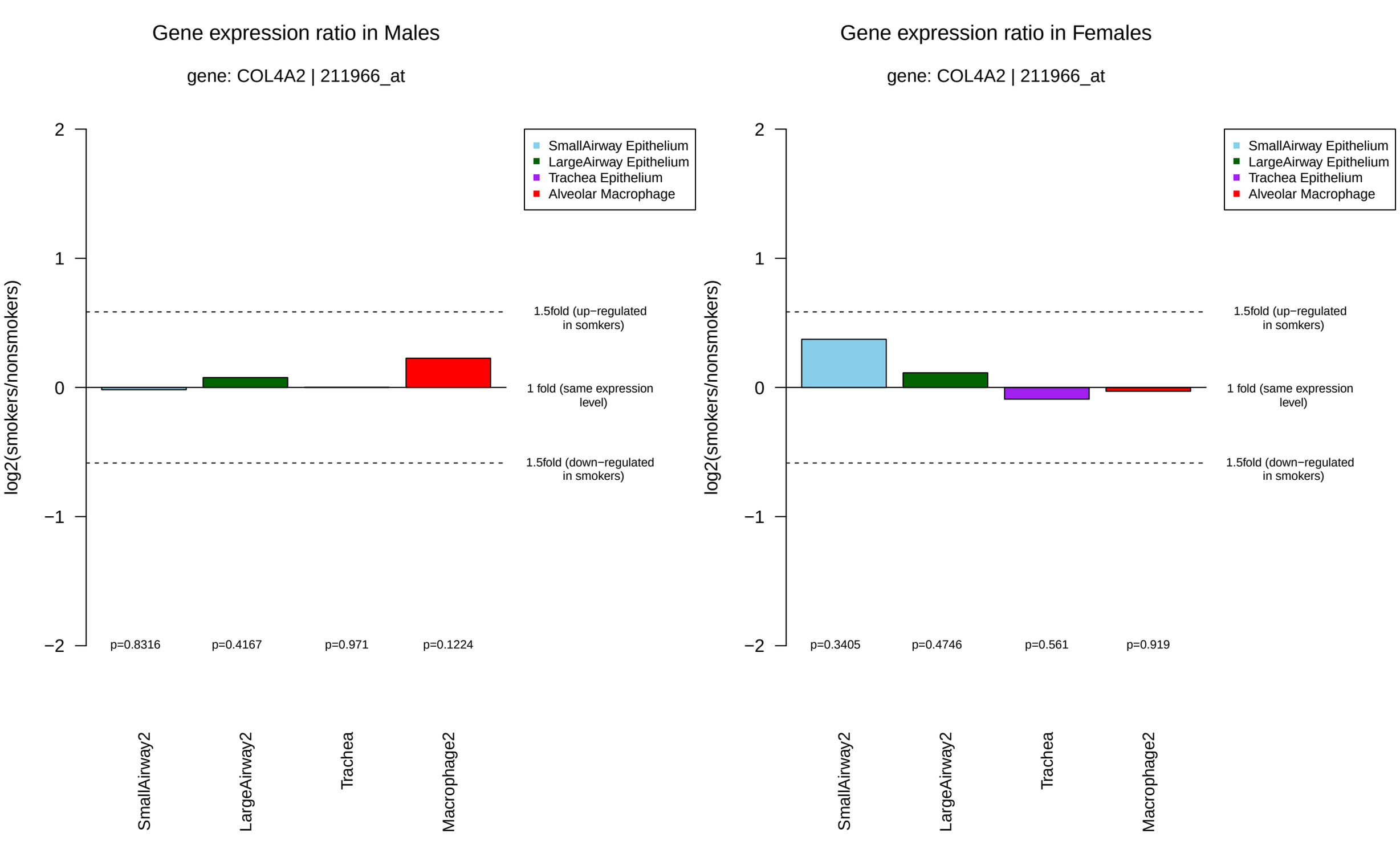
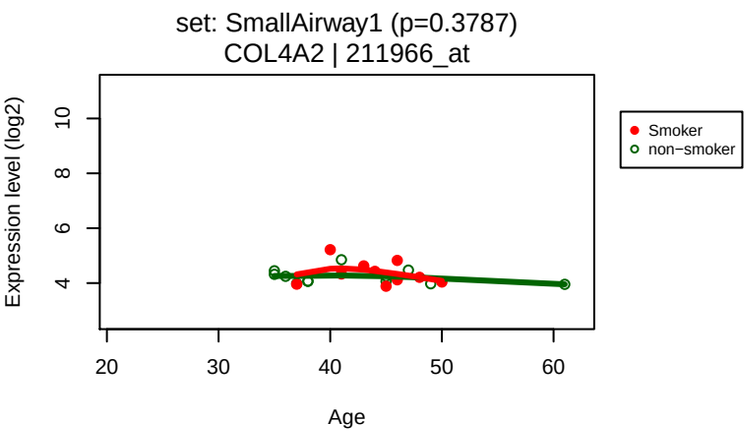
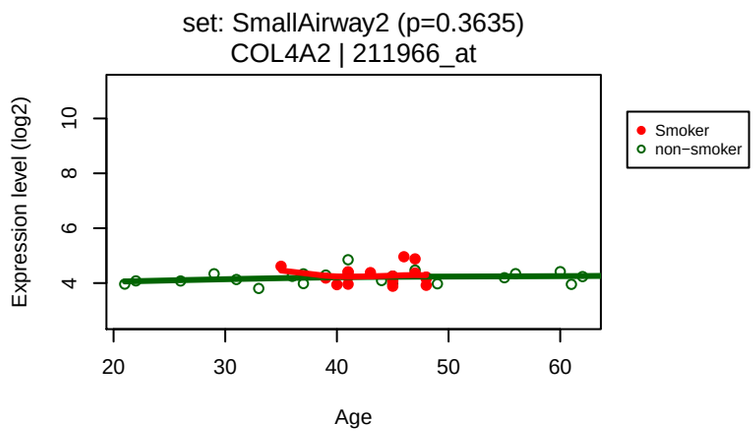
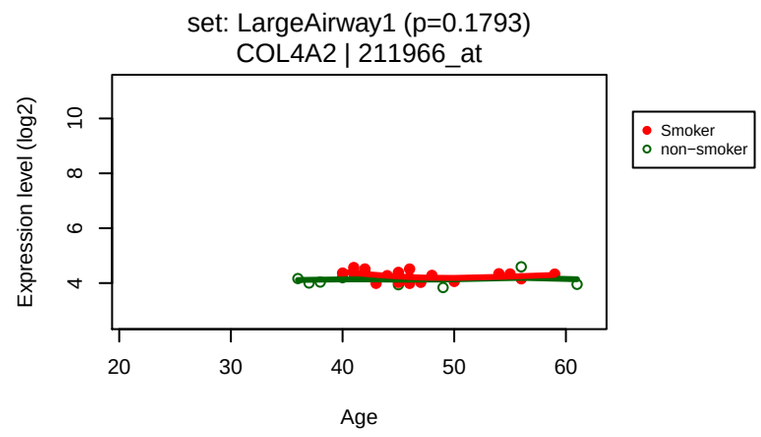

Gene expression of smokers and nonsmokers
gene: COL4A2 | 211966_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3787
p=0.3635
p=0.1598
p=0.1793
p=0.1769
p=0.7512
p=0.9054
p=0.2548
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

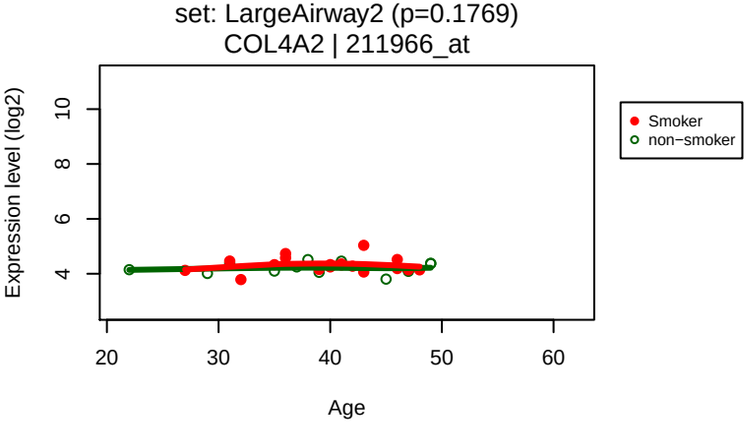

Gene expression ratio (smokers/nonsmokers)
gene: COL4A2 | 211966_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3787
p=0.3635
p=0.1598
p=0.1793
p=0.1769
p=0.7512
p=0.9054
p=0.2548
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

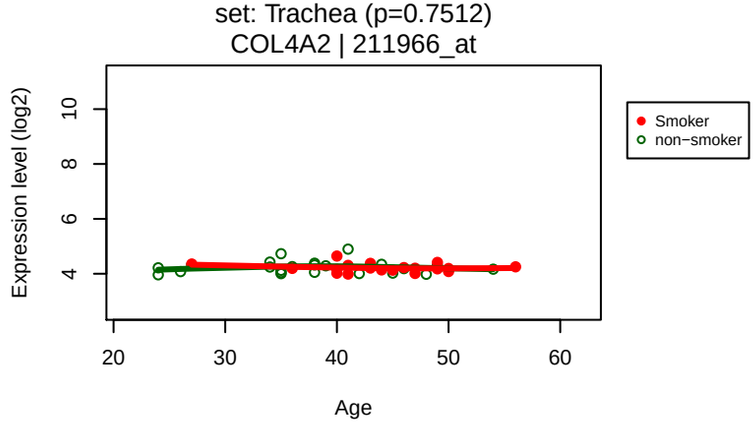

Gene expression ratio in Males
gene: COL4A2 | 211966_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8316
p=0.4167
p=0.971
p=0.1224
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A2 | 211966_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3405
p=0.4746
p=0.561
p=0.919
SmallAirway2
LargeAirway2
Trachea
Macrophage2

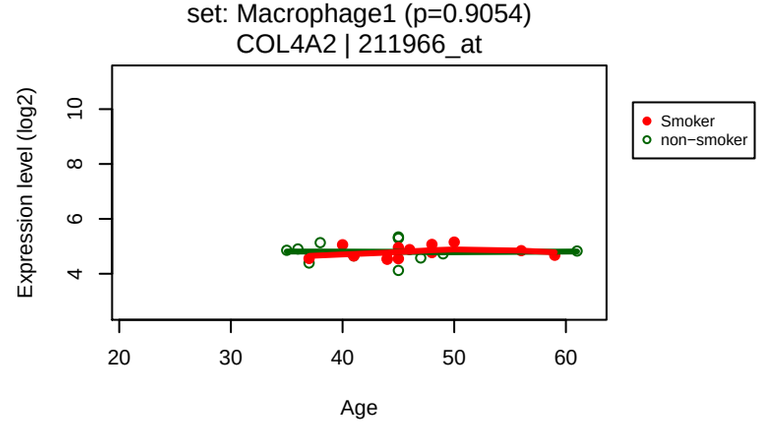

set: SmallAirway1 (p=0.3787)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker

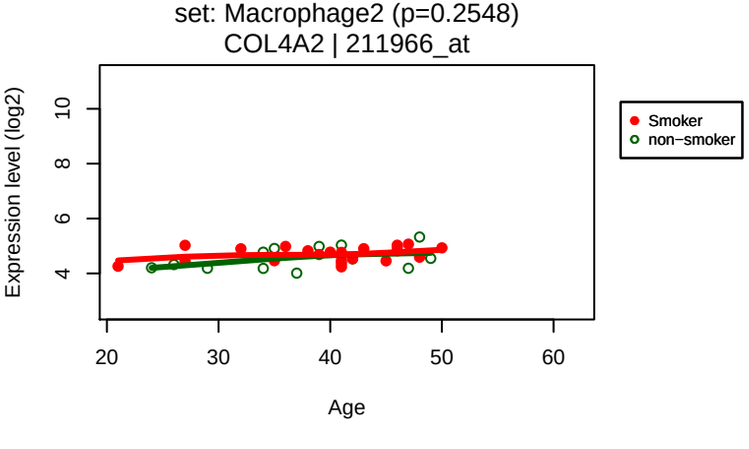

set: SmallAirway2 (p=0.3635)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker

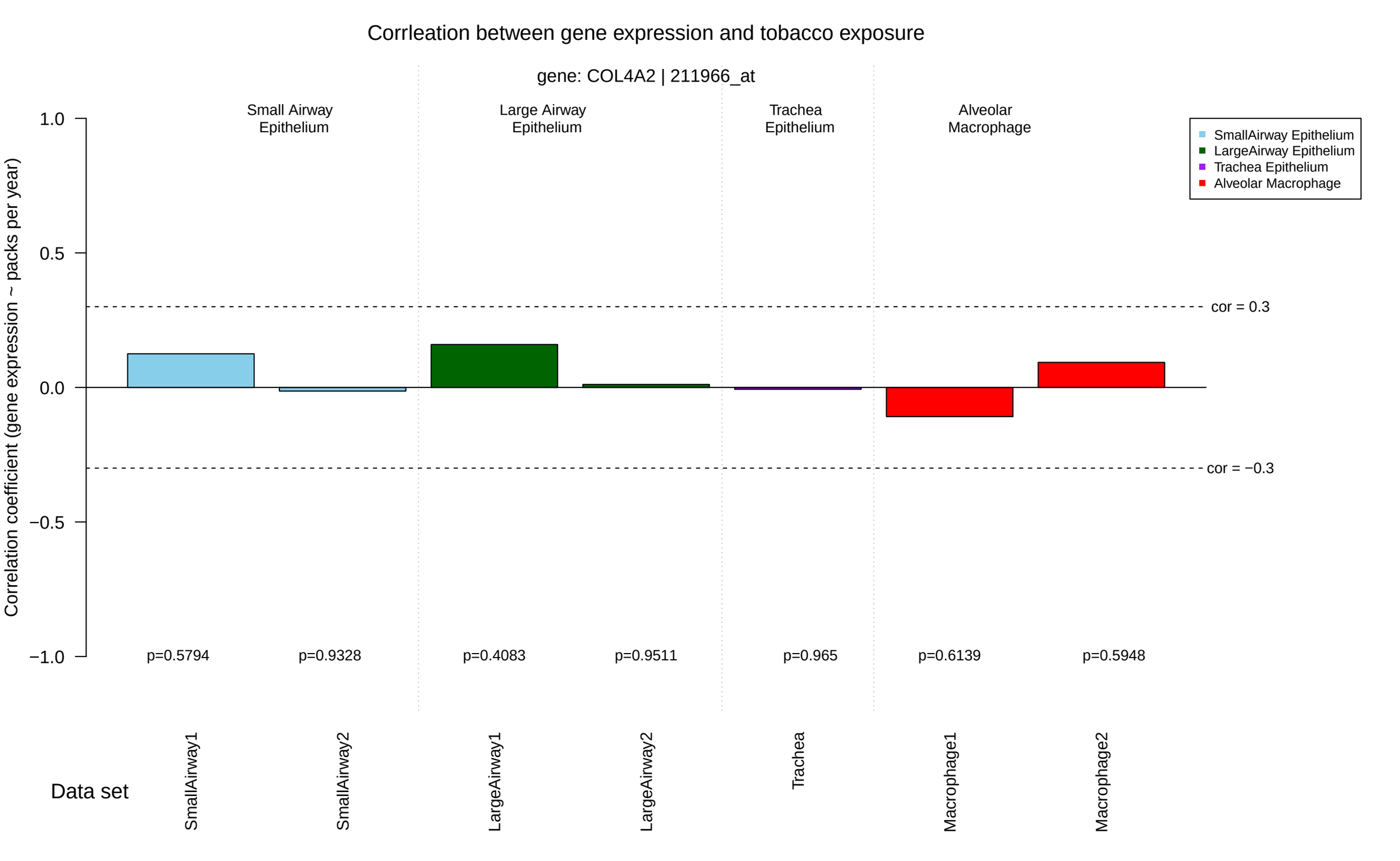

set: LargeAirway1 (p=0.1793)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1769)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7512)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9054)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2548)
COL4A2 | 211966_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A2 | 211966_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5794
p=0.9328
p=0.4083
p=0.9511
p=0.965
p=0.6139
p=0.5948
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2